Computer Networks

→ What is Computer ?

⇸ A computer is an electronic device that :
- Takes input
- Processes the input
- Stores data
- Gives output

Eg: · You type a message on whatsapp.
· Your computer/mobile processes it.
· It sends the message to your friend.

· But a single computer can only use its own data. What if we want to share data with another computer?

This is where Computer Networks Come in!

→ What does "Network" mean ?

⇸ The word network come from two words :
- Net: A web or interconnected structure.
- Work: To function or operate together

So, a network means:
A group of connected thing that work together.

A network doesn't have to involve computers. The concept exist everywhere.

Eg- 1: Road Network:-

City A-------- City B

City D----------- City C

All ~~city~~ cities are connected by roads.

This is called a road network.

Eg- 2 :- Railway Network :-

Purnea ----Patna ---Kolkata

Kanpur --- Varanasi

Many railway stations are connected by railway tracks.

This is a railway network.

Eg :- Social network :-

Akash------ Ayush

Aishanya---- Vivek

Instagram, linked, and whatsapp connect people.

Date:

Eg-4 :- Mobile Network :-

When we call some one :

Our phone
¦
Mobile tower
¦
Service provider
¦
Mobile tower
¦
friend's phone

All these devices are connected
This is a mobile network.

Eg-5 :- Computer Network :-

Computer A-------Computer B
| ¦
Computer D-------Computer C

Now all the computers are connected.
This is called a computer network.

· Networking is the process of connecting computers so they can exchange information.
Just like roads connect two city for which people can travel, just like that data can travel from computer to another.

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→ Computer Network :-

A computer network is a system of interconnect devices, such as computers, servers, smartphones, and printers, that communicates to exchange data.

They communicates using wired or wireless connections and can range from small home network to the global ~~network~~ internet.

· What does "Communication" mean?

Communication means:

Sending and recieving information.

In networking,

Computer A

↓

message

↓

Computer B

→ Basic components of communication :-

Every communication needs five things:

1.) Sender :

Who sends the message.

Eg;

I send a whatsapp message

I am the sender.

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2.) Reciever:

Who recieves the message.

My friend is the reciever.

3.) Message:

The information being sent.

Ex:-

- Image
- Video
- text
- PDF

4.) Transmission Medium:

The path through which the message travels.

Eg:-

- Wi-Fi
- optical fiber
- Ethernet cable
- Mobile Network

5.) Protocol:-

The rules of communication.

Just like traffic follows traffic rules,
Computer follow communication rules called protocols.
Without protocol, computer cannot understand each other.

Eg: Person A → Hindi
Person B → Chinese
→ Communication becomes difficult
Both need common rules or a common language.

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→ What is data?

Data means raw facts.

Ex:

Akash 21 97

These are data

After processing:

- Akash is 21 years old and scored 97 marks.

This become information.

→ Information:-

Processed and meaningful data is called information.

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→ Why do we need Networking ?
let's think about life before networks existed.

We have two computers :-

Computer A   Computer B

Computer A has :
- Notes
- Movies
- Songs
- Internet
- Printer

Computer B has nothing.

Now suppose Computer B wants the notes.

Q. How will it get them?
Without a network, you only options are :
- USB drive
- CD / DVD
- External hard disk

This is :
- Slow
- Inconvenient
- Time-Consuming

So people thought :
"Why don't we connect computers directly ?"

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thats how networking begin's.

→ The main purpose of Networking :

To connect computers so they can communicate and share resource efficiently.

Before roads :

Village A   village B
No direct connection

After roads :-

Village A ===== Village B

people and goods ~~ca~~ can travel easily.

Similarly,

Computer A ==== Computer B

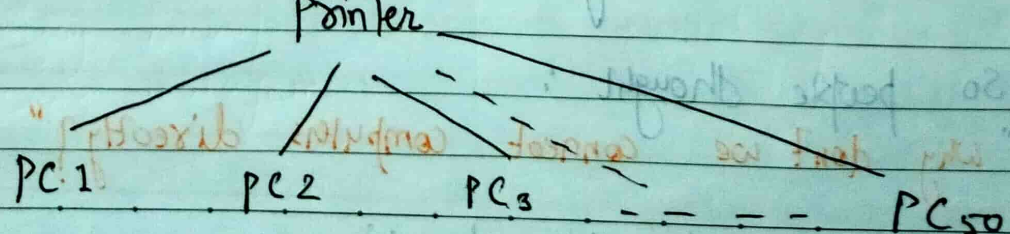
Data can travel

→ Why is Networking Needed?

1.) Resource Sharing :-

Suppose an office has 50 Computers.
should they buy 50 printers?

No.



All computers share one printer.

Similarly, the can share:

- Scanner
- Storage
- Software
- Internet

This saves Money & time.

2.4 File sharing :-

We are making a project.

Akash creates:

Akash-project.pdf

Rahul also needs it.

Without networking :

- Copy to USB
- Give the USB
- Copy again.

With networking :

Akash PC --------- Rahul PC

Just send the file in seconds.

Date:

3.) Communication :-

Networking allows people to communicate instantly.

Eg)

- Whatsapp
- Email
- Zoom
- Google meet
- Microsoft teams

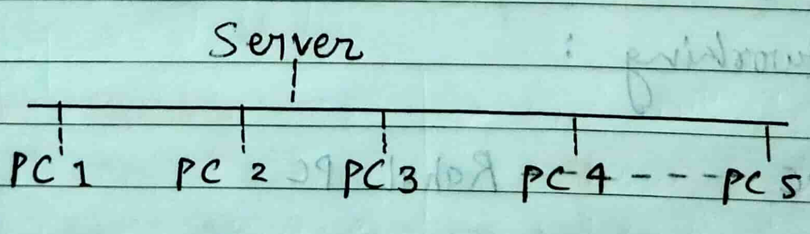
- Without networking, none of these would exist.

4.) Data sharing :-

Banks have many branches.
Suppose I ~~you~~ deposit money in Delhi.
Next day, ~~you~~ I withdraw it in patna.
How does the patna branch know your balance?
Because all branches are connected through a network.

5.) Centralized data :-

Instead of storing data on every computer, a company stores everything on one server.



Benifits:-

- Easy backup
- Easier management

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6.) Cost Reduction :-

Imagine 100 computers.

Without networking:
- 100 printers
- 100 internet connections

With networking:
- Shared printer
- Shared internet

7.) Faster Communication :-

Earlier:
Write a letter:
3-5 days

Now:
WhatsApp message
Seconds

Networking makes communication almost instantaneous

8.) Online Services :-

Many modern services depend on networking:
- Online shopping
- UPI payments
- ATM transactions
- Railway ticket booking
- Flight booking
- Video streaming
- Online gaming

Without networking, these services wouldn't work.

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9) Remote Access :-

Suppose I am at home

My office computer is in office, Delhi.

Using networking:

Home laptop
|
internet
|
office computer

You can access files remotely.
This is widely used in work-from-home environments.

10.) Internet Access :-

The internet itself is a massive network.

When we search:
"what is computer network?"
My request goes from:

Laptop
↓
Wi-Fi
↓
Router
↓
Internet
↓
Google Server
↓
back to us

→ We will learn later how it happen in detail. Our subject is known for how networking work.

→ Data flow :-

There are two computers.

Computer A ========== Computer B

Now,

How will the data move?

→ There are three possibilities :

1.) Data moves only from A to B.
2.) Data moves both ways, but one at a time.
3.) Data moves both ways at the same time.

These three possibilities are called Data flow modes.

Data flow :-

Data flow is the direction in which data travels between two communicating devices.

→ Types of data flow :-

There are three types :-

Data flow
- Simplex
- Half Duplex
- Full Duplex

1) Simplex Communication:-

In simplex communication, data flows in only one direction.

One device is always the sender

The other device is always the reciever

Sender ------> Reciever

The reciever cannot send data back.

Eg; ① Television

TV station ------> MY TV

The TV station sends signals
MY TV only recieve them

It cannot send anything back.

② Radio

Radio Station ---> Radio

Only one-way communication.

→ Advantages:-
① Simple to implement
② Low cost
③ Fast for one-way communication

→ Disadvantages:
- No feedback
- Reciever cannot respond
- Not suitable for interactive communication

2.4 Half Duplex Communication:-

In half Duplex communication, data can flow in both directions, but not at the same time.

Both devices can send and recieve.

However,
Only one device can transmit at a time.

Computer A ←-----→ Computer B
(Not simultaneously)

Eg; Walkie - Talkie {use radio-waves to communicate}
{Range usually → 1 – 10 km}

Two Army officers use walkie-Talkies.

officer A:
"over"

officer B wait

Then replies
"Roger"

But can speak.
But not simultaneously.

Date:

A ---> B

or

B ----> A

Never both together

→ Advantages:-
- Two-way communication
- less expensive than full duplex

→ Disadvantages:-
- Waiting time
- lower efficiency
- Slower than full duplex

3.) Full duplex communication:-
In Full duplex communication, data flows in both direction simultaneously.

Both devices can :
- send data
- Recieve data

at the same time

Computer A <======> Computer B

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Eg ; Mobile phone calls

vedio calls

→ Advantages :-
- fast Communication
- No waiting
- Efficient
- Better performance

→ Disadvantage :-
- More expensive
- More complex hardware.

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→ Types of computer network :-

A computer network is a group of two or more independent computers and devices connected to share data, resources.
These ~~technology~~ connections can be established using wired media or wireless technology.

Types of computer networks are classified based on several factors:

1. Based on purpose & area covered
2. Based on types of communication
3. Based on type of Architecture

1.) Based on purpose & area covered :-
Categorizing computer networks according to the physical distance or area they cover, purpose is to ranging from small personal space to large global regions.

Based on geographical area, computer networks are mainly divided into five types:

(i) Personal Area Network (PAN) :-
This connects personal devices within a very short range around a single user.

- Coverage Range: 1 - 10 meter
- Eg;

Blue tooth :

Wireless earbuds
↑ Bluetooth
↓
Mobile

→ PAN have limited Bandwidth, so they are not suitable for large scale.

Bandwidth → It means the max^m amount of data that a communication link can carry per unit time.

Eg; USB Dongle :

· It commonly expressed as bits per sec. bps, Kbps, Mbps, Gbps etc.

Wireless mouse (emit Radio signal (2.4 GHz))

↓

USB Dongle

↓

computer / laptop

The Dongle recieves the wireless signal and passes the I/P to the computer.

When it is used:

For short-range communication b/w personal devices.

(ii) Local Area Network (LAN) :-

This connect computer and devices within a Limited area such as home, office, school, college, or building. It usually uses switches, routers.

· Coverage Range: One room, building, or small campus.

Eg; Wi-fi & Ethernet cable

Eg → College computer lab has 30 computer

Internet

|

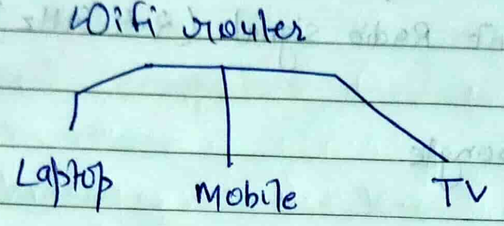
Router

|

Switch

PC1 PC2 PC3 PC4

Eg: Home network :



Eg: Wired LAN :-

PC → Ethernet cable → switch.

When it is used :
Fast and secure communication within a local location.

(iii) Campus Area Network (CAN) :-
A CAN connects multiple LANs within a campus or a group of nearby buildings.

- Coverage ~~Campus~~ Range :-

University or cooperate campus or big hospitals

Area covered : Typically 1 to 10 km.

Eg; Computer A [let CSE building]

Ethernet Cable

switch

— optical fibre —

switch

Computer B [EEE building]

When it is used :-

To interconnect LANs across a campus under one organization.

(iv) Metropolitan Area Network (MAN) :-

A MAN connects multiple network within a city or metropolitan region :

- Coverage Range : City or large town
- Key - technologies : Fibre optics, microwave, metro ethernet

LAN+LAN+LAN

↓

MAN

- When it is used :

To provide high-speed connectivity across a city.

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(V) Wide Area Network (WAN) :-

A WAN connects networks over large geographical area such as countries or continents.

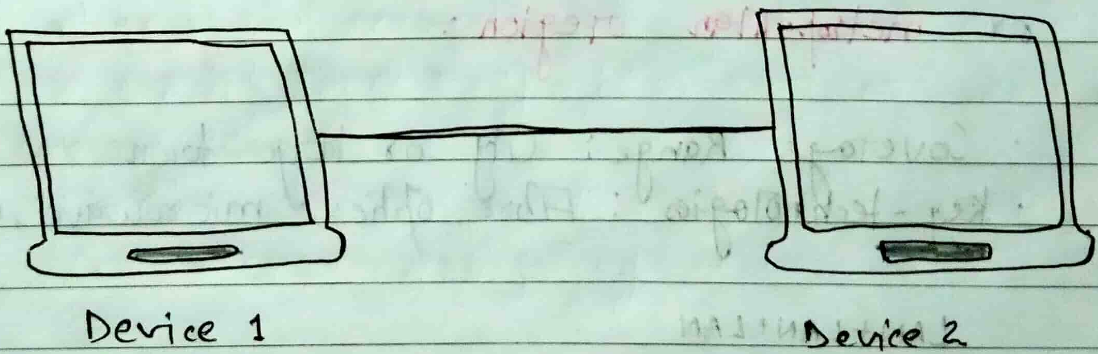
- Coverage area: Country, continent or global
- Key technologies: Satellite links, Internet.

When it is used: To enable long-distance and global communication.

2.4 Based on types of communication :-

Network can also be divided based on the types of communication they use:

(i) Point-to-point :-



There is a dedicated communication path b/w the two devices.

Eg; • Mobile phone call
• Sending a file from one laptop to another
•

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· Advantages :-

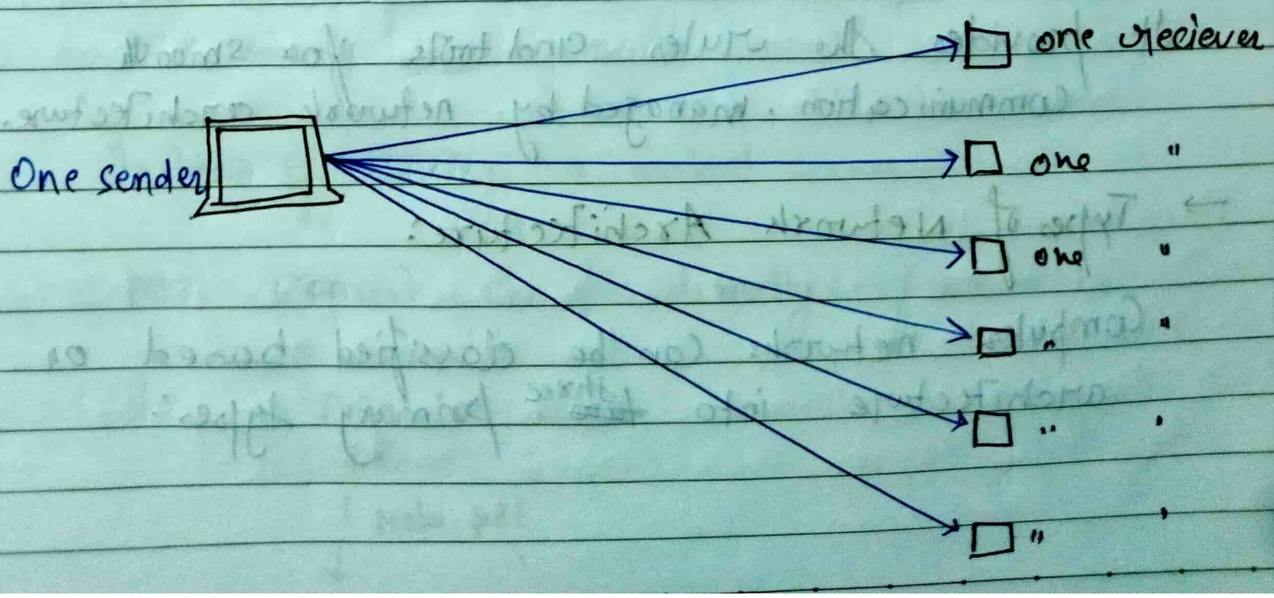
· High security
· provide a dedicated and direct connection, ensuring reliable communication b/w two devices.

· Disadvantages :-
· Not suitable for communicating with many devices simultaneously

(ii) Boardcast Network :-
In a Boardcast Network, one sender send data to all devices on the network simultaneously.
In this setup, data travels in one direction, from the sender to all potential recievers.

Eg ;
Radio station, where the station broadcasts signals that can be picked up by the radio reciever within range.

Eg; Television network broadcasting to multiple viewers.



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→ Advantages:-

- Efficient for distributing data to a large number of recipients simultaneously.
- Simple to implement for one-to-many communication scenarios.
- Reduces the need for individual connections to each reciever, saving resources.

→ Disadvantages:-

- Data is ~~sent~~ sent in one direction, limiting two-way communication.
- Can lead to bandwidth wastage if not all recievers need the transmitted data.

3.) Based on Type of architecture :-

Network Architecture is the design of a computer or communication network that shows how devices, software, protocols, and media work together.

It defines the physical and logical structure, task allocation, and connectivity between client like laptops and servers.

- It provides the rules and tools for smooth communication, managed by network architecture.

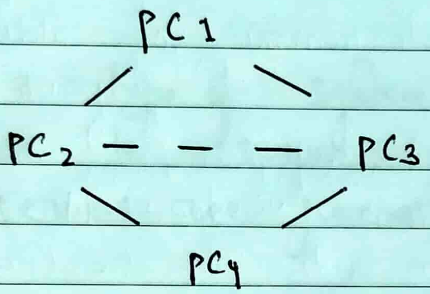
→ Types of Network Architecture:

- Computer networks can be classified based on architecture into ~~two~~ three primary types.

(i) Peer-to-peer architecture:- (P2P network):-

A peer-to-peer (P2P) network is a network in which all computers have equal status. Every computer can acts as both a client and a server.

There is no central server.



Each computer communicates directly with the others.

Working:

There are two computers:

PC1 — — — — — — — PC2

PC1 has a file called csenotes.pdf
PC2 wants that file.

Instead of asking a central server,

PC2 request the file directly from PC1.

PC1
|
| Notes.pdf
↓
PC2

here:
- PC1 acts as a server (sharing a file)
- PC2 acts as a client (requesting the file).

later, if PC1 needs a movie from PC2:

$$PC2 \xrightarrow{\text{movie.mp4}} PC1$$

Now,
PC2 becomes the server and PC1 becomes the client.
This is why it is called peer-to-peer every computer is equal.

→ Advantages :-
- P2P network is less costly and cheaper, it is affordable.
- P2P is very simple and not complex.
  This is because all computers that are connected in network communication in an efficient and well-mannered with each other.

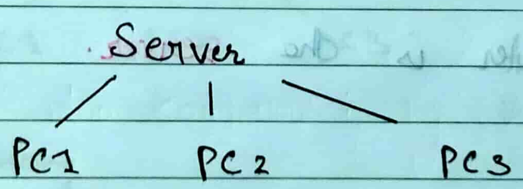
→ Disadvantages :-
- It is more difficult to manage security policies
- Difficult to manage as the network grows.
- Performance decreases with many computers.

(ii) Client - Server Network :-

CSN (client - server network) is a type of computer network in which one of the centralized and powerful computers (commonly known as server) is hub to which many of personal computer that are less powerful or workstations (commonly known as client) are connected.

It is a type of system where ~~clien~~ clients are connected to server to just share or use resources. These servers are generally considered as heart of system.

This type of network is more stable and scalable as compared to P2P network.



These server stores:

- files
- Databases
- Applications
- User accounts

Client request service from the server.

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Eg;

When we open www.google.com :

My laptop
↓ Request
Google server
↓
Search result
↓
My laptop

My laptop is ~~[illegible]~~ the client.

Google's computer is the server.

→ Advantages:
- Dedicated server
- Centralized management
- High security
- Suitable for large organizations.
- Easy maintenance.
- Easy backups

→ Disadvantages:
- Expensive
- Requires a dedicated server
- If the server fails, client may lose access to services.

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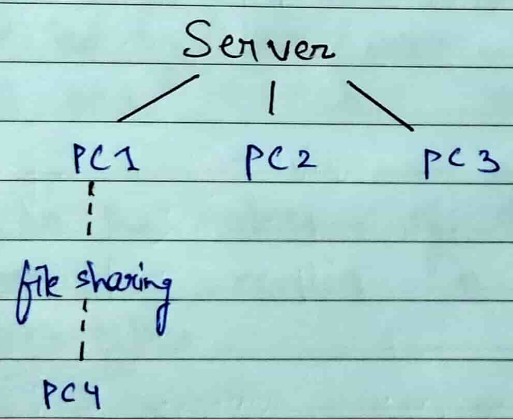
Eg;

- Banks
- Colleges Libraries
- Google
- facebook etc.

(iii) Hybrid network :-

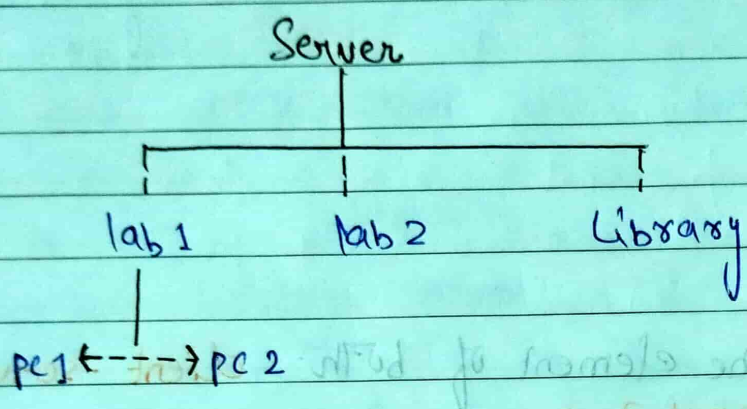
Hybrid network combine element of both client-server peer-to-peer architecture.

Some computers communicate directly with each other (peer-to-peer), while others use a central server.



- here, 1. PC1, PC2, and PC3 access the central server for files and databases.
  2. PC1 and PC4 can also share files directly with each other.

Eg: In a college:



· Students access the college server.
· Computers inside a lab may also share files directly.

→ Transmission media in Computer networks :-

Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network.

It forms the fundamental pathway through which information is transmitted, ensuring connectivity between networked devices.

Data travels through the transmission medium.

→ The selection of transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (Bandwidth)
- Affected by interference and Noise
- Cost and installation requirements.

Based on the nature of the transmission path, transmission media is broadly classified into two main types:

1.) Guided Media :-

Guided media also known as wired or bounded transmission media, refers to transmission media in which data signals are transmitted through a physical path using cables.

The signal is confined and guided along a fixed route, providing controlled communication b/w network devices.

Computer ---cable---- Computer

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Types of guided media:-

(i) Twisted pair cable :-

A twisted pair cable consists of two insulated copper wires twisted together.

Twisting reduces electromagnetic interference (EMI).

Types :-

(a) UTP [Unshielded twisted pair) :-

UTP cables consists of twisted ~~of~~ copper wire pairs without any additional metallic ~~sto~~ shielding.

~~Advantages~~ Property:

- No shielding
- Cheap
- flexible

Eg;

- Home LAN
- Office LAN
- Ethernet (Electrical signal

(b) Shielded Twisted pair (TSP) :-

STP cable consists of twisted pair of insulated copper conductors enclosed within an additional metallic shielding layer, such as foil, or braided ~~shield~~ ~~shied~~. shield.

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property:
- Extra metal shielding
- Better protection against noise
- More expensive

(ii) Coaxial cable:-

A coaxial cable has a central copper conductor surrounded by insulation, metallic shielding, and an outer protective jacket.

Uses:-
- Cable TV
- CCTV

Advantages:
- Better noise resistance than twisted pair
- longer transmission distance
- Higher bandwidth

Disadvantages:
- More expensive than twisted pair
- Thicker and less flexible.

(iii) Optical fibre cable:-

An optical fibre cable transmits data using light signal instead of electrical signals.

Advantages:-
- Very high speed
- Very high bandwidth
- long distance communication
- High security

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· Disadvantages:
- Expensive
- Difficult to install

Uses:
- Internet backbone
- jio fibre
- Undersea cable
- Data centers

2.) Unguided media:

Unguided media, also known as wireless or Unbounded transmission media, uses electromagnetic waves to transmit data without any physical medium.
Data travels through air or ~~free~~ free space using electromagnetic waves.
No physical cable is required.

characteristics:
- No cable
- Easy installation
- Convienient.

Type of Unguided media:

(i) Radio waves: frequency range: 3KHz - 300GHz
Uses:
- FM radio
- Bluetooth
- Wi-fi
- Mobile communication

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Components:-

- Transmitter: Generates and modulates the signal.
- Reciever: Recieves and demodulates the signal.

(ii) Microwaves:-

Microwaves uses line-of-sight communication, where the transmitting and recieving antennas must be properly aligned. The ~~a~~ transmission range depends on the height of the antennas.

frequency range: 1 GHz – 300 GHz.

Uses:-

- Mobile towers
- Satellite communication
- Long-distance communication

features:-

- High speed
- long distance

(iii) Infrared :-

~~Infrared~~ Infrared waves are used for short-range wireless communication and operates in the frequency range of 300 GH to 400 THZ.

Uses:-

- TV remote
- Air conditioner
- Short-range devices

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(iv) Satellite communication :-

Uses :-

- GPS
- Satellite TV
- Weather forecasting
- Global communication

Advantages :-

- Very large coverage
- Global connective.

Disadvantage :-

- Higher cost

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→ Types of Network topology :-

Network topology :-
Network topology is the physical or logical arrangement of computers (node) and communication links (cables or wireless connections) in a computer network.

In simple terms,
Network topology tells us how computers are connected to each other in a network.

→ Why do we need network topology?
Suppose we have 5 computers.
Ques:
How should they be connected?
There are many possible ways.

PC1 ---- PC2 ----- PC3 ----- PC4 ----- PC5

or;

```
            PC1
             |
PC2 ---- PC3 ---- PC4
             |
            PC5
```

Each arrangement has different :
- Cost
- performance
- Reliability
- Ease of maintenance

Date:

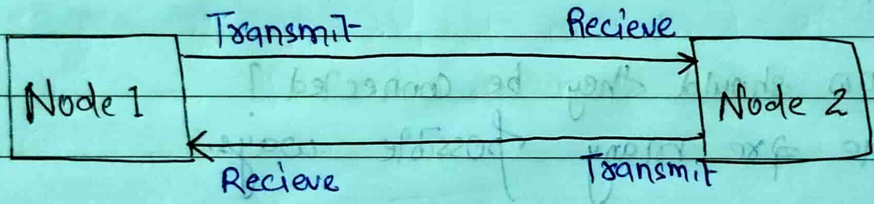
Types of topology :-

(i) Point-to-point topology :-

Point-to-point topology is a type of topology that works on the functionality of the sender and reciever.

It is the simplest communication between two nodes, in which one is the sender and the other one is the reciever.

It provides high bandwidth.



Advantages :-

- Simple & easy to setup
- High data transfer speed
- Secure communication

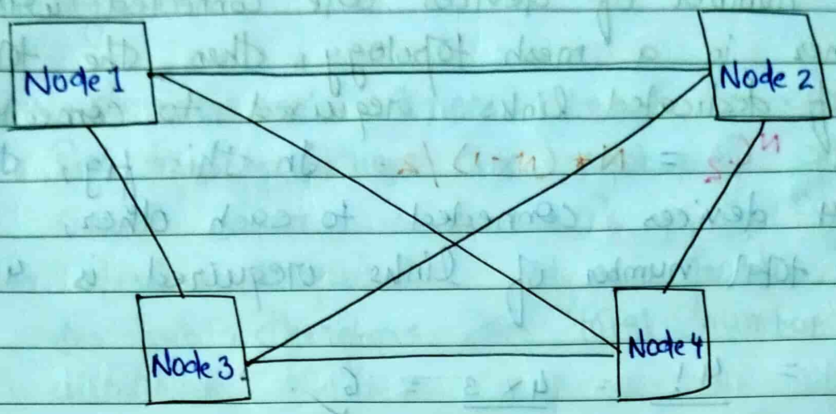
Disadvantages :-

- Not suitable for large networks.
- Limited scalability.

(ii) Mesh topology :-

In mesh topology, every computer is connected to every other computer.

Ports → Physical connection point on a device where a network cables can be connected.



· features :-
  · Multiple communication paths
  · very reliable

· Advantages :-
  · High reliability
  · High security
  · Failure of one link usually does not stop communication.

· Disadvantages :-
  · Very expensive
  · large amount of cabling
  · Complex installation.

Eg; Suppose, the N number of device are connected with each other in a mesh topology, the total number of ports ~~required~~ that are required by each device is N - 1.
In the given fig, there are 4 devices connected to each other, hence total number of ports required by each devices is 3. The total no. of ports required $= N * (N-1)$.

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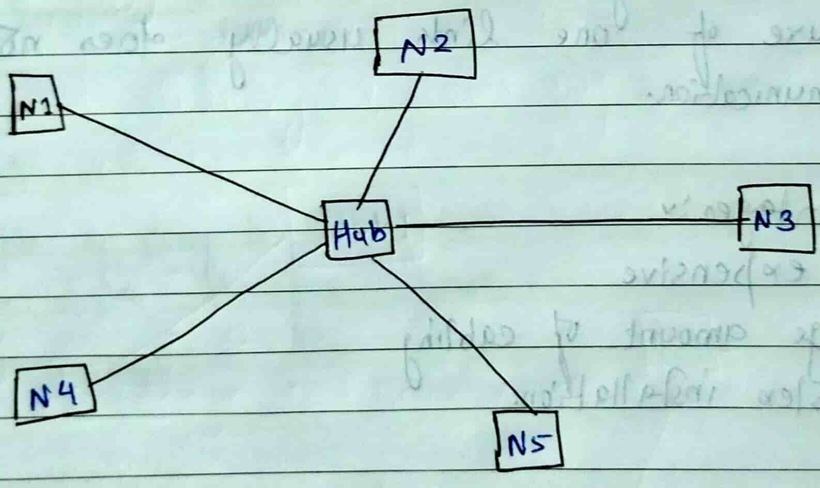
- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connected them is $^NC_2 = N*(N-1)/2$. In this fig, there are 4 devices connected to each other, hence total number of links required is $^4C_2$

$$= \frac{4!}{2! \times 2!} = \frac{4 \times 3}{2} = 6$$

(iii) Star topology:-

In star topology, all the devices are connected to a single hub through a cable.
This hub is the central node and all other nodes are connected to the central node.



Features:-

- Every computers has its own connection to the center.
- Easy to add or remove computers.

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Advantages:-

- If N devices are connected ~~are~~ to each other in a star topology, then the number of cables required to connect them is N. So it is easy to set up.
- Each devices requireds only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.
- Easy to fault identification and fault isolation.

Disadvantages:-

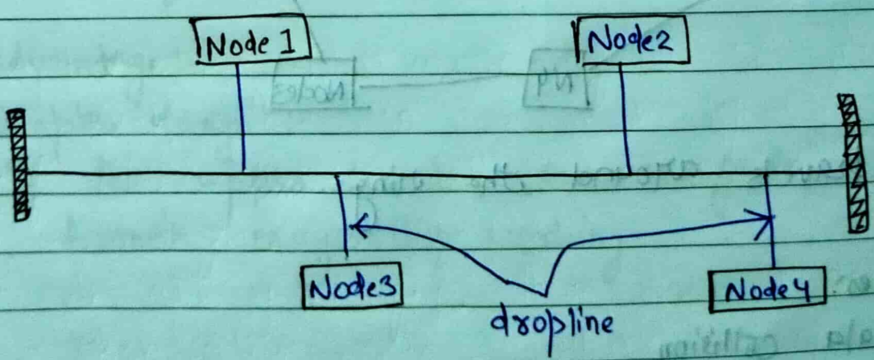
- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- Performance is based on the single concentrator i.e. hub.

(iv) Bus topology:-

Bus topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional.

In simple terms,

All computers are connected to a single main cable, called the backbone cable.



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→ Advantages:

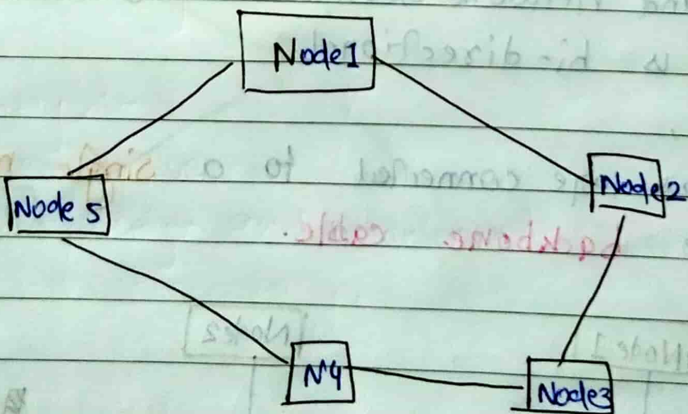
- If N devices are connected to each other in a bus, then the number of cables required to connected them is 1, known as backbone cable, and N drop lines required

→ Disadvantages:-

- A bus topology is quiet simpler, but still, it requires a lot of cabling.
- If the common cables fails, then the whole system will crash down.
- Adding new devices to the network would slow down networks.
- Security is very low.

(V) Ring Topology :-

Each computer is connected to two neighbouring computers, forming a closed loop.



Data travels around the ring.

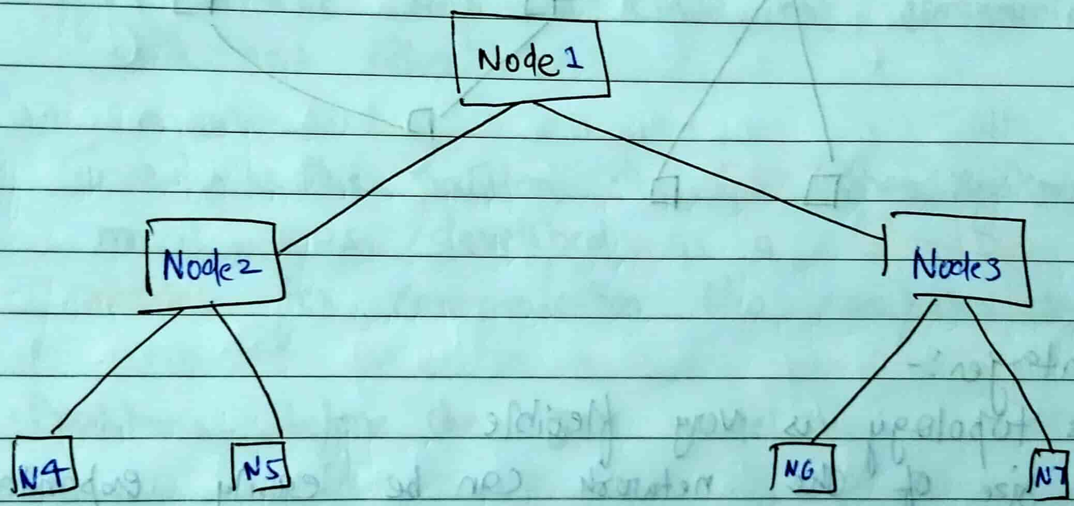
Advantages:

- No data collision
- Equal access for all computers.

disadvantages:-
- Failure of one cable or node can affect the network.
- Difficult to expand.
- less secure

(vi) Tree topology:-

A tree topology combines multiple level in a hierarical structure.



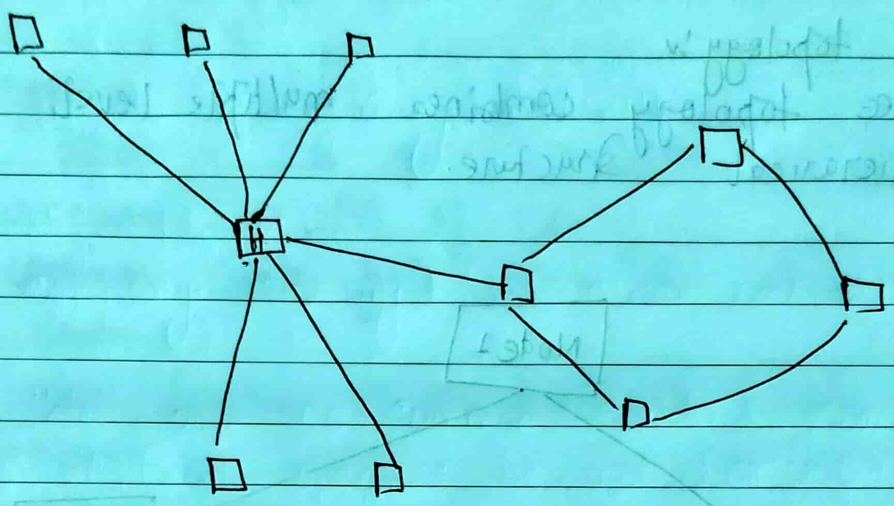
Advantages:
- Easy to expand
- Easy management
- Suitable for large organization.

Disadvantage:-
- Complex design.
- If the upper-level connection fails, the affected branch may stop working

Akash

(vii) Hybrid Topology:-

Hybrid topology is the combination of all the various types of topology we have studied above.



Advantages:-

- This topology is very flexible
- The size of the network can be easily expanded by adding new devices.

Disadvantages:-

- Very expensive
- Complex design and maintenance.